

1) Solve the following matrix multiplication problems

$$\begin{bmatrix} 8 & 9 & 3 \\ 3 & 2 & 2 \end{bmatrix} * \begin{bmatrix} 5 & 4 \\ 1 & 2 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 58 & 56 \\ 23 & 20 \end{bmatrix}$$

$\begin{matrix} 2 \times 3 \\ 3 \times 2 \end{matrix}$

$8*5+9*1+3*3=58$ You multiply the first row, first column
 $8*4+9*2+3*2=56$ You multiply the first row, second column
 $3*5+2*1+2*3=23$ You multiply the second row, first column
 $3*4+2*2+2*2=20$ You multiply the second row, second column

$$\begin{bmatrix} 8 \\ 2 \end{bmatrix} * \begin{bmatrix} 8 & 5 \\ 7 & 6 \\ 9 & 1 \end{bmatrix} = \text{Not solvable}$$

$\begin{matrix} 2 \times 1 \\ 3 \times 2 \end{matrix}$ Not conformable

$$\begin{bmatrix} 2 & 1 & 4 \end{bmatrix} * \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \end{bmatrix}$$

$\begin{matrix} 1 \times 3 \\ 3 \times 1 \end{matrix}$

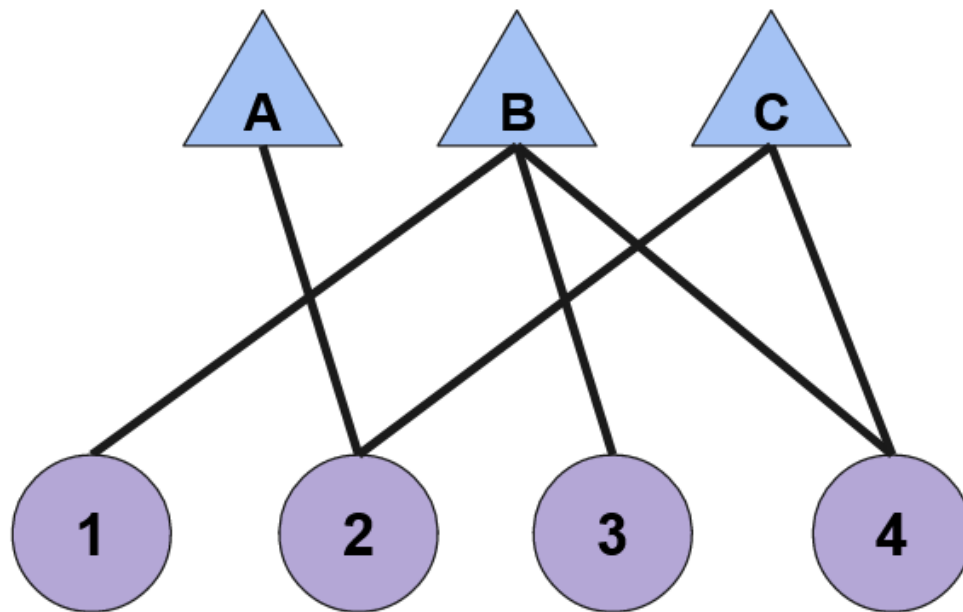
$2*3+1*3+4*0=9$ You multiply the first row, first column

$$\begin{bmatrix} 5 & 5 \\ 3 & 2 \\ 3 & 1 \end{bmatrix} * \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 25 \\ 13 \\ 11 \end{bmatrix}$$

$\begin{matrix} 3 \times 2 \\ 2 \times 1 \end{matrix}$

$5*3+5*2=25$ You multiply the first row, first column
 $3*3+2*2=13$ You multiply the second row, first column
 $3*3+1*2=11$ You multiply the third row, first column

2) Affiliation Network



a) What is the affiliation matrix of the above graph?

	A	B	C
1	0	1	0
2	1	0	1
3	0	1	0
4	0	1	1

The adjacency is determined by a tie between the member and the group. So 1 is not a member of A (0), but is a member of B (1).

b) What is the member overlap matrix for the above affiliation matrix?

	A	B	C
1	0	1	0
2	1	0	1
3	0	1	0
4	0	1	1

4x3

×

	1	2	3	4
A	0	1	0	0
B	1	0	1	1
C	0	1	0	1

3x4

The adjacency matrix

The adjacency matrix transpose

Rows become columns, columns become rows

	1	2	3	4
1	1	0	1	1
2	0	2	0	1
3	1	0	1	1
4	1	1	1	2

Multiply each of the rows and columns as done in part 1.

3) What do you do according to the theory?

a. Tom is a rude person. According to karma theory you?

You would feel negatively towards Tom. Karma theory states that you will receive the valences you send out.

b. Everyone likes Aruna. According to status theory you?

Also like Aruna. Status theory says people will match their valences to what other people send towards someone. So if other people like someone, the ego/you will too.

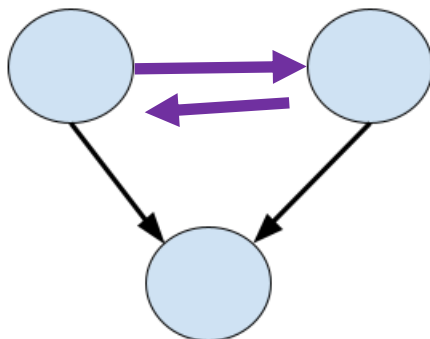
c. No one likes Christophe. According to solidarity theory you?

Depends on how people feel about you. Solidarity theory says that we feel towards someone depending on the mutual position in the network. So if other people also don't like you, then you will like Christophe because you are mutually disliked, but if other people like you, then you won't like Christophe because you do not share the same social position.

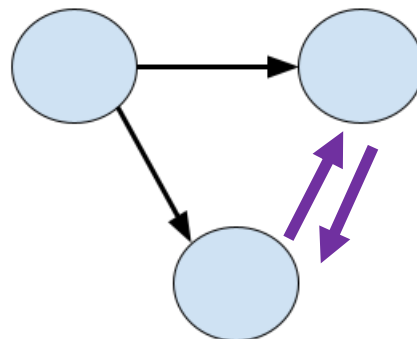
d. Manuel is a friendly person who's nice to all. According to karma theory you?

You like Manuel. Karma theory states that you will receive the valences you send out.

4) The following two triads are hierarchies. First, do winner, loser, and/or by-stander effects create the dominance order for (a) and (b)? Close the triad by making the hierarchy transitive. Justify the direction of the transitivity. Is the example of an attribute or interaction-based model?



(a)



(b)

For graph a, the loser effect creates the graph, where the two nodes send dominance to the bottom node. This is because the bottom node competed first, and lost, meaning that it then lost the subsequent contest due to loser effects, such as injury, lost confidence, etc.

For the graph b, the winner effect creates the graph, as the top left node sends dominance to the other two nodes. This would happen if the top left node engaged in a contest first, and won, and then went on to win the next contest due to winner effects, such as increased confidence, experience, etc.

It actually doesn't matter which way we draw the arrows to make the graph transitive, it just needs to be 1 arrow where one is currently missing (like the purple ones). Depending on which way you draw the arrow though changes which way the order of hierarchical competition occurred though. Whichever node sends the arrow is the node that didn't engage in the first competition, but was rather the bystander. Whichever node receives a purple arrow is the original loser of the competition, and has gone on to lose again due to loser effects. As such, this graph is depicting the interaction-based model of hierarchy formation.

6. Definitions

Two-Mode- A two-mode **network** is composed of two different types of nodes-members and their membership categories. A two-mode **projection** is a **graph** composed of two different types of nodes-members and their membership categories.

Affiliation Network- Same thing as a two-mode network, consists of the affiliations between members and membership categories.

Bipartite Graph- Same thing as a two-mode projection. A graph that depicts the relationship between members and membership categories.

Affiliation Matrix- A matrix that depicts the relationships of affiliation networks, connects members on the rows i and membership categories on columns j .

Matrix Transpose- The inversion of a matrix by transforming rows into columns and columns into rows.

Co-Membership Matrix- A matrix that results when multiplying the affiliation matrix by its transpose, results in the overlap of how many groups each member shares with another member.

Group Overlap Matrix- A matrix that results when multiplying the affiliation matrix transpose by the original affiliation matrix, results in the overlap of how many members each group shares with another group.